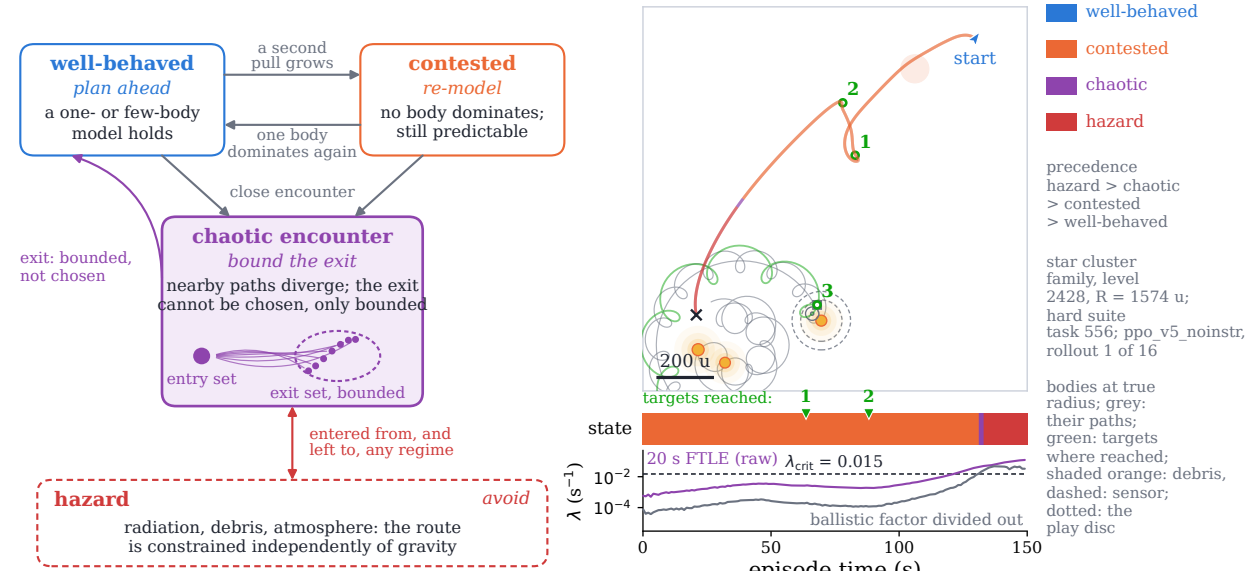

(a) regimes as a state machine
well-behaved
plan ahead
a one- or few-body model holds
a second pull grows
contested
re-model
no body dominates; still predictable
one body dominates again
close encounter
chaotic encounter
bound the exit
nearby paths diverge; the exit cannot be chosen, only bounded
entry set
exit set, bounded
exit: bounded, not chosen
entered from, and left to, any regime
hazard
avoid
radiation, debris, atmosphere: the route is constrained independently of gravity
(b) one flight, read through it
start
1
2
3
200 u
targets reached:
1
2
state
20 s FTLE (raw)
λ_{crit} = 0.015
10^{-2}
10^{-4}
λ (s^{-1})
ballistic factor divided out
0
50
100
150
episode time (s)
well-behaved
contested
chaotic
hazard
precedence
hazard > chaotic
> contested
> well-behaved
star cluster
family, level
2428, R = 1574 u;
hard suite
task 556; ppo_v5_noinstr,
rollout 1 of 16
bodies at true
radius; grey:
their paths;
green: targets
where reached;
shaded orange: debris,
dashed: sensor;
dotted: the
play disc